ORGANIC CHEMISTRY FRONTIERS

CCS 中国化学会 CHINESE CHEMICAL SOCIETY

SIOC 1950

ROYAL SOCIETY OF CHEMISTRY

RESEARCH ARTICLE

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Org. Chem. Front.*, 2021, **8**, 3390

Selectivity between an Alder–ene reaction and a [2 + 2] cycloaddition in the intramolecular reactions of allene-tethered arynes†

Anh Le and Daesung Lee *

Received 25th March 2021,
Accepted 15th April 2021

DOI: 10.1039/d1qo00459j

rsc.li/frontiers-organic

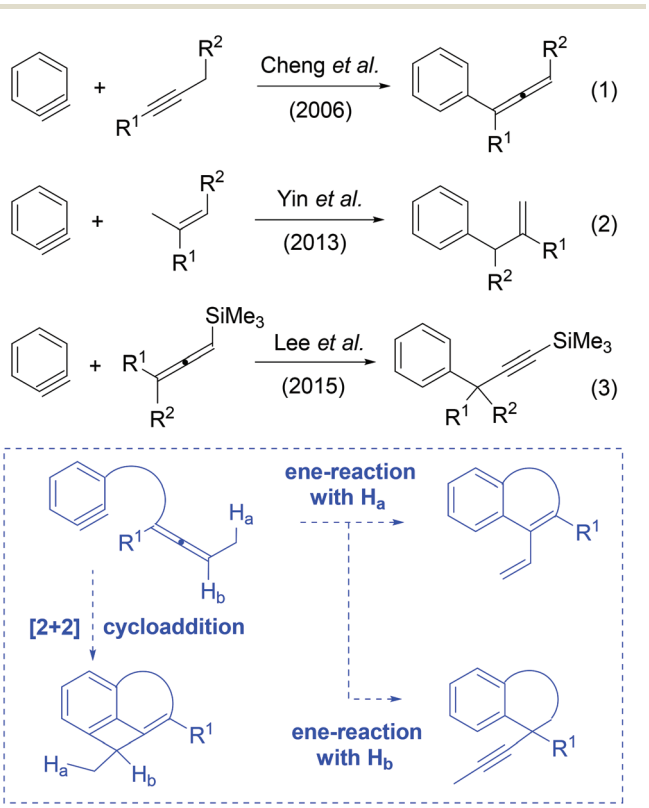
Substituent-dependent reactivity and selectivity in the intramolecular reactions of arynes tethered with an allene are described. With a 1,3-disubstituted allene moiety, an Alder–ene reaction of an allenic C–H bond is preferred over a [2 + 2] cycloaddition, whereas a [2 + 2] cycloaddition of the terminal π-bond of the allene is preferred with a 1,1-disubstituted allene. With a 1,1,3-trisubstituted allene-tethered aryne, an Alder–ene reaction with an allylic C–H bond is preferred over a [2 + 2] cycloaddition.

Published on 15 April 2021. Downloaded by Fudan University on 1/18/2022 2:29:23 AM.

A wide variety of electron-deficient π-systems have been employed as an ene-acceptor under either thermal or Lewis acid-catalyzed conditions.[1] Due to the electrophilic nature of benzyne species,[2] they can behave as an efficient ene-acceptor[3] to react with ene-donors containing π-systems such as alkynes, alkenes, and allenes (Scheme 1). In 2006, Cheng reported intermolecular Alder–ene reactions between benzyne and terminal and internal alkynes (eqn (1)),[4] and the corresponding reaction with alkenes was also reported by Yin in 2013 (eqn (2)),[5] which was further extended to alkenes containing a polar functional group[6] by employing arynes generated from tetraynes *via* the hexadehydro Diels–Alder reaction.[7] Recently, Lee and coworkers also explored the Alder ene reaction between benzyne and silylallenes (eqn (3)).[8]

In 2011, Lautens and coworkers reported intramolecular Alder–ene reactions of arynes generated *via* a strong base-mediated elimination with aryl bromides.[9] Recently, Hoye[10] and Lee[11] reported the intramolecular Alder–ene reactions of arynes generated from tri- and tetraynes under thermal conditions. At this juncture, we want to further explore the scope and selectivity of the intramolecular Alder–ene reactions by accommodating an allene as the ene-donor (eqn (4)).[12] The main concern in this intramolecular reaction would be the Alder–ene selectivity between the allylic and allenic C–H bonds, which is expected to mainly depend on the substituent pattern of the allene moiety. In this communication, we describe the reactivity and selectivity trend of intramolecular Alder–ene reactions that also compete with the [2 + 2] cycloaddition of the terminal π-system of the allene.[13]

First, we examined the Alder–ene reaction by using symmetrical tetraynes **1a–1c** (Scheme 2). The reaction of tetrayne **1a** that contains a three-atom tether with a *gem*-dimethylated allene[14] moiety (toluene, 90 °C, 8 h) afforded the 7-membered



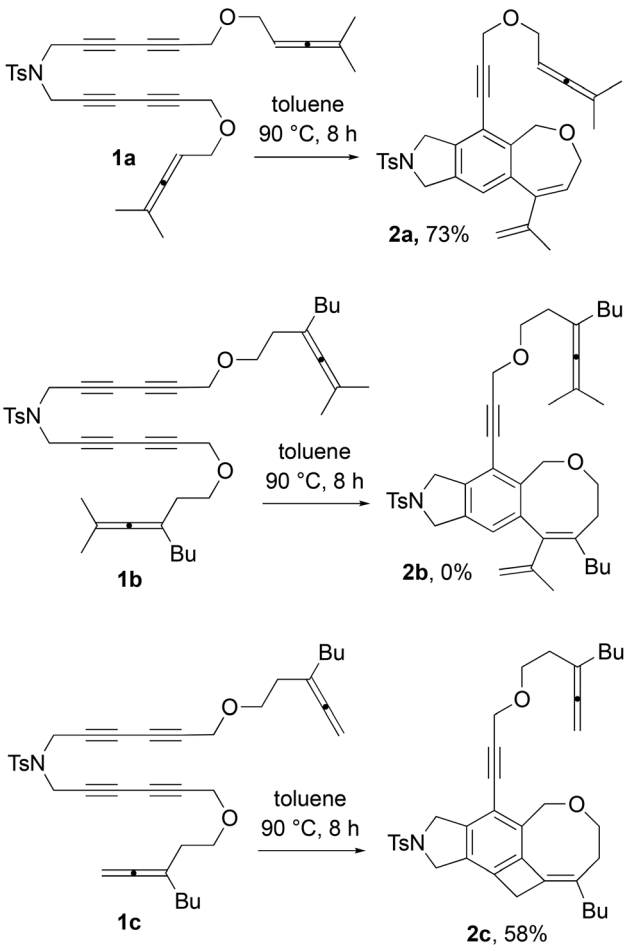
Scheme 1 Alder–ene reactions of arynes.

Department of Chemistry, University of Illinois at Chicago, Chicago, Illinois, USA. E-mail: dsunglee@uic.edu

† Electronic supplementary information (ESI) available: Detailed experimental procedure, ^{1}H and ^{13}C spectra. See DOI: 10.1039/d1qo00459j

This journal is © the Partner Organisations 2021

Published on 15 April 2021. Downloaded by Fudan University on 1/18/2022 2:29:23 AM.



Scheme 2 Reactions of symmetrical tetraynes tethered with an allene moiety.

ring Alder–ene product **2a** in 73% yield. On the other hand, tetrayne **1b** containing a four-atom tether with a tetrasubstituted allene moiety did not afford either the Type-I or Type-II ene reaction product, instead, the decomposition of **1b** was observed. Under identical conditions, however, substrate **1c** containing a terminal allene moiety afforded only the [2 + 2] cycloadduct engaging the terminal π-bond of the allene[15] to generate **2c** in 58% yield, and the Alder–ene product was not observed.

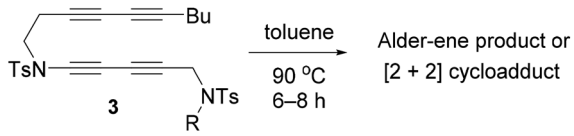
On the basis of the drastic change of the reaction modes depending on the substituent pattern in allenes, we further explored the general trend of the reactivity of unsymmetrical tetraynes **3a–3h** that contain structurally different allene moieties (Table 1). As expected, tetrayne **3a** bearing a three-atom tether with a *gem*-dimethyl allene moiety exclusively formed the 7-membered Alder–ene product **4a** in 63% yield (entry 1). In stark contrast, tetrayne **3b** containing a 1,3-disubstituted allene[16] moiety results in the decomposition of the starting material and neither Alder–ene product **4b** nor **4b′** of the allenic or allylic C–H bond was observed (entry 2). Substrate **3c** containing a 1,1-disubstituted terminal allene also decomposed without providing either the Type-I or Type-II ene product **4c** or **4c′** (entry 3). Surprisingly, however, substrates **3d** and **3d′** containing a 1,3-disubtituted allene with an extra methylene exclusively engaged in the Alder–ene reaction with an allenic C–H bond[8] to afford **4d** and **4d′** in 64 and 60% yield, respectively (entries 4 and 5), and the Type-1 ene product of the corresponding allylic C–H bond was not observed. The reaction of the mono-substituted allene in **3e** induced the Type-I ene reaction of an allenic C–H bond to provide the terminal alkyne-containing product **4e** in 55% yield (entry 6). Substrate **3f** containing a 1,1-disubstituted terminal allene, which is identical to **3e** but contains a triethylsilyl group at the internal position of the allene, afforded the [2 + 2] cycloadduct **4f** in 62% yield and no Alder–ene product was observed (entry 7). Both substrates **3g** and **3h** containing a 1,1,3- or 1,3,3-trisubstituted allene[17] moiety, respectively, afforded the corresponding Type-I Alder–ene products **4g** and **4h** in 34 and 58% yield, respectively (entries 4 and 5).

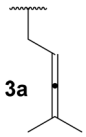
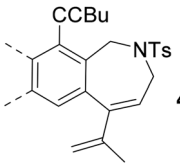
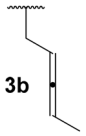
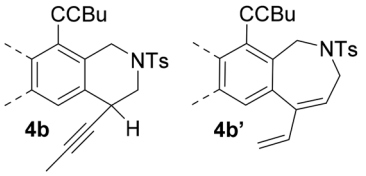
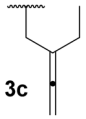
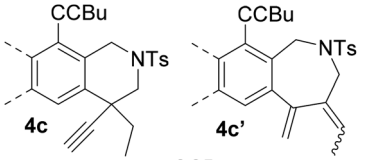
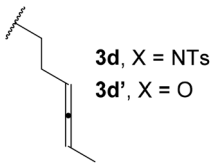
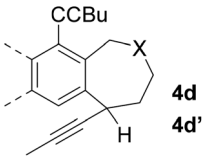
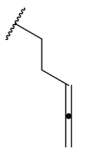
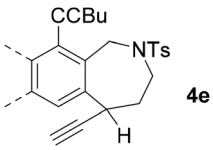
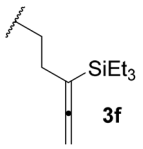
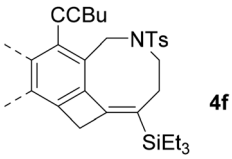
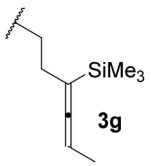
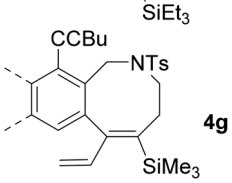
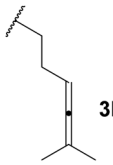
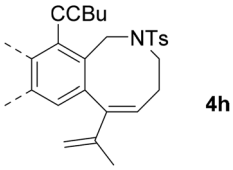
These examples suggest that the preference for the formation of different modes of Alder–ene or [2 + 2] cycloaddition depends subtly on the substituents on the allene moiety and on the length of the tether between the allene and the aryne. In general, the Alder–ene reaction of an allylic C–H bond is most favorable with trisubstituted allenes (entries 1, 8 and 9), whereas that of an allenic C–H bond becomes more favorable with monosubstituted and 1,3-disubstitued allenes (entries 4–6). Alternatively, the 1,1-disubstituted terminal allene induces [2 + 2] cycloaddition over an Alder–ene reaction (entry 7).

Next, we explored the reactivity of arynes generated from triynes **5a–5f** containing different allene moieties (Table 2). The reaction of triynes **5a** and **5b** bearing a *gem*-dimethyl-containing trisubstituted allene or a 1,3-disubstituted allene moiety led to only decomposition, without generating the expected 7-membered ring Type-I Alder–ene products **6a** and **6b** (entries 1 and 2). On the other hand, triynes **5c** with a longer tether bearing a 1,3-disubstituted allene moiety afforded the Type-I ene product **6c** in 72% yield, where only an allenic C–H bond participated in the reaction (entry 3). As expected, substrate **5d** bearing a *gem*-dimethyl-containing trisubstituted allene afforded an 8-membered ring Type-I Alder–ene product **6d** in 50% yield (entry 4). It is quite surprising to find that **5e** containing a 1,1-disubstituted terminal allene moiety did not participate in the expected [2 + 2] cycloaddition between the allene and aryne, instead the toluene moiety of the NTs group participated in a Diels–Alder reaction to generate benzobarrelene[18] **6e**, where the allene moiety remained intact (entry 5). The preference of a Diels–Alder reaction is further demonstrated with triyne **5f** devoid of the allene moiety, which provided benzobarrelene **6f** in 58% yield (entry 6). Despite the identical allene moieties in tetraynes **3f** and triynes **5e**, their reaction outcomes are quite different, which strongly suggests that the reactivity of the putative aryne intermediates is intricately affected by multiple factors including the substituents on the aryne core moiety. In comparison, a *gem*-dimethyl allene-containing propiolated triynes **5g** exclusively afforded the Alder–ene reaction product **6g** in marginal yield (entry 7). The formation of product **6g** suggests that the

Published on 15 April 2021. Downloaded by Fudan University on 1/18/2022 2:29:23 AM.

Table 1 Reactions of ynamide-tethered tetraynes tethered to different allene moieties



Entries	R	Product	Yield[a] (%)
1	 **3a**	 **4a**	63
2	 **3b**	 **4b** **4b'**	0[b,c]
3	 **3c**	 **4c** **4c'**	0[b,c]
4	 **3d**, X = NTs **3d'**, X = O	 **4d** **4d'**	64
5			60
6	 **3e**	 **4e**	55
7	 **3f**	 **4f**	62
8	 **3g**	 **4g**	34
9	 **3h**	 **4h**	58

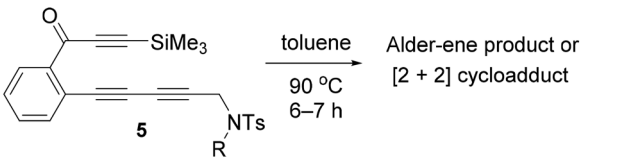
[a] Isolated yield. [b] Decomposition of the starting material. [c] Decomposition of the starting material in toluene at 55 °C with 5 mol% of $AgSbF_6$.

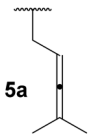
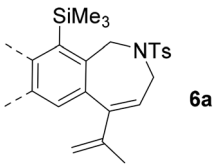
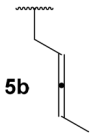
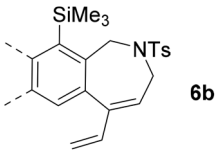
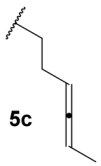
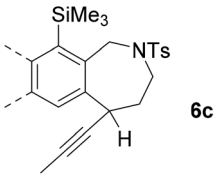
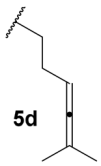
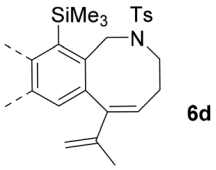
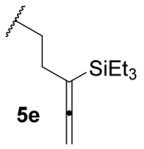
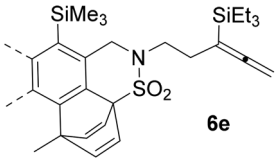
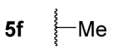
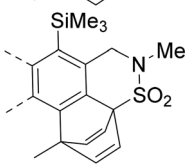
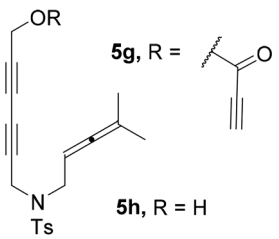
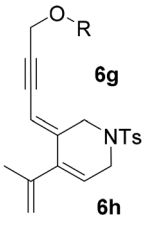
hexadehydro Diels–Alder reaction of **5g** requires a higher activation barrier than the Alder–ene reaction between the alkyne moiety and the tethered allene segment. It was assumed that the low yield from this reaction is the consequence of the instability of the propiolate ester functionality at the elevated temperature. Indeed, when 1,3-diynyl propargyl alcohol **5h** was subjected to identical conditions, the corresponding Alder–ene product **6h** was obtained in 62% yield (entry 8).

 This journal is © the Partner Organisations 2021

Published on 15 April 2021. Downloaded by Fudan University on 1/18/2022 2:29:23 AM.

Table 2 Reactivity of benzene-tethered triynes containing different allene moieties

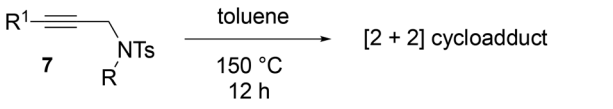


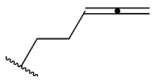
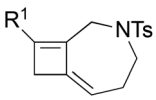
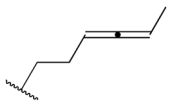
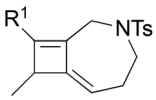
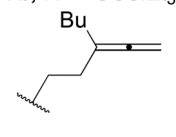
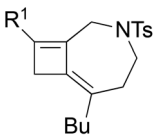
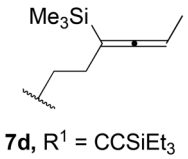
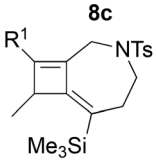
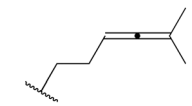
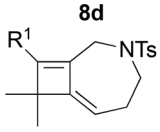
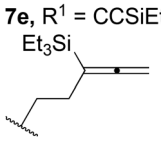
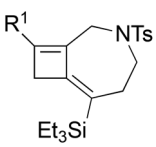
Entries	R	Product	Yield[a] (%)
1	 **5a**	 **6a**	0[b,c]
2	 **5b**	 **6b**	0[b,c]
3	 **5c**	 **6c**	72
4	 **5d**	 **6d**	50
5	 **5e**	 **6e**	34
6	 **5f**	 **6f**	58
7	 **5g**, R =	 **6g**	32[d]
8	**5h**, R = H	**6h**	62[d]

[a] Isolated yield. [b] Decomposition of the starting material. [c] Decomposition of the starting material in toluene at 55 °C with 5 mol% of $AgSbF_6$. [d] Reaction at 150 °C for 12 h.

Having recognized the significant impact of the structure of the tether, alkynes and allene moiety, we employed various substrates that contain a 1,3-diyne (**7a–7f**) and monoynes

Table 3 Reactivity of the sulfoamide-tethered alkyne containing different allene moieties



Entries	R	Product	Yield[a] (%)
1	 **7a**, R^1 = $CCSiEt_3$	 **8a**	82
2	 **7b**, R^1 = $CCSiEt_3$	 **8b**	85
3	 **7c**, R^1 = $CCSiEt_3$	 **8c**	80
4	 **7d**, R^1 = $CCSiEt_3$	 **8d**	78
5	 **7e**, R^1 = $CCSiEt_3$	 **8e**	65
6			78
7	**7ga**, R^1 = H	**8ga**	0[b]
8	**7gb**, R^1 = Br	**8gb**	0[b]
9	**7gc**, R^1 = Bu	**8gc**	0[b]
10	**7gd**, R^1 = Ph	**8gc**	0[b]

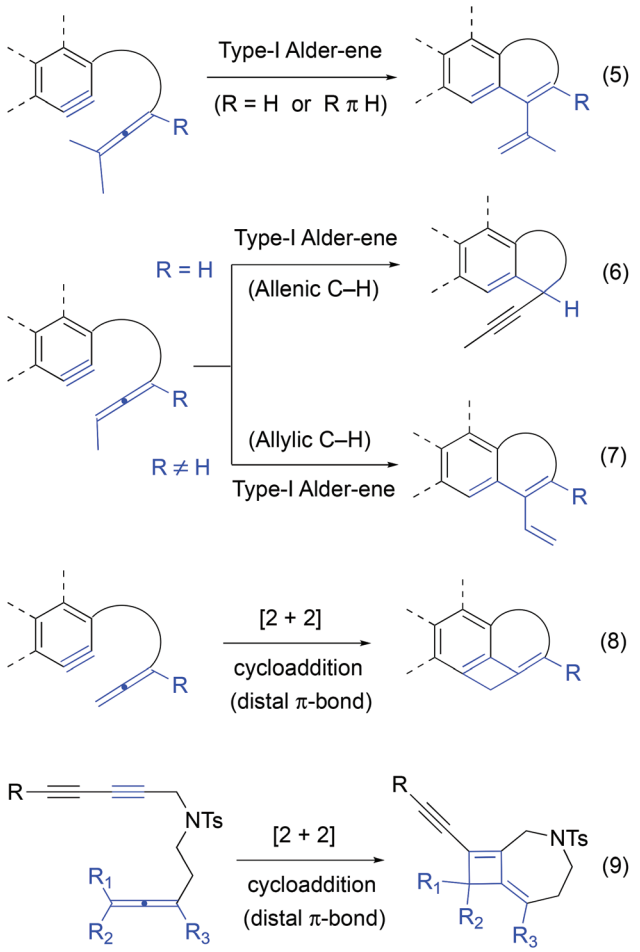
[a] Isolated yield. [b] Decomposition of the starting material.

(**7ga–7gd**) tethered with differently substituted allene moieties to further explore the selectivity between the Alder–ene and a [2 + 2] cycloaddition (Table 3). Upon heating (150 °C, toluene, 12 h), all 1,3-diyne-tethered allenes **7a–7f** afforded [2 + 2] cycloadducts **8a–8f** in good yields (entries 1–6), whereas monoynes **7ga–7gd** only led to decomposition under identical conditions (entries 7–10) and none of the expected [2 + 2] cycloadducts **8ga–8gd** were observed.

The activating role of the extra alkynyl substituent at the terminal position of the alkyne in these [2 + 2] cycloadditions is noteworthy.[19]

In summary, we have systematically investigated the intramolecular reaction between allenes and arynes by employing

Published on 15 April 2021. Downloaded by Fudan University on 1/18/2022 2:29:23 AM.



Scheme 3 A general trend of selectivity in the intramolecular reaction of allenes with an aryne.

allene-tethered tetraynes and triynes as aryne precursors. From the data accumulated in Scheme 2 and Tables 1–3, a general reactivity and selectivity trend has emerged (Scheme 3). Allenes containing the *gem*-dimethyl group at the distal carbon exclusively participate in the Type-I ene reaction regardless of the substituent at the proximal carbon (eqn (5)). 1,3-Disubstituted allene favorably participates in the Alder–ene reaction with an allenic C–H bond (eqn (6)), whereas 1,1,3-trisubstituted allene prefers to generate the Alder–ene with an allylic-H bond (eqn (7)). The reaction between a 1,1-disubstituted terminal allene and an aryne favors a [2 + 2] cycloaddition (eqn (8)). On the other hand, the reaction between 1,3-diyne and an allene moiety results in a [2 + 2] cycloaddition product irrespective of the substituent pattern of the allene (eqn (9)). This general reactivity trend would be a useful guide for further investigations of aryne chemistry involving allene counterparts.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

We are thankful for the financial support from NSF (CHE-1764141) and the mass spectrometry facility at UIUC.

Notes and references

1 For general reviews on ene reactions, see: (*a*) H. M. R. Hoffmann, The Ene Reaction, *Angew. Chem., Int. Ed. Engl.*, 1969, **8**, 556; (*b*) B. B. Snider, Lewis-acid Catalyzed Ene Reactions, *Acc. Chem. Res.*, 1980, **13**, 426; (*c*) K. Mikami and M. Shimizu, Asymmetric Ene Reactions in Organic Synthesis, *Chem. Rev.*, 1992, **92**, 1021; (*d*) L. C. Dias, Chiral Lewis Acid Catalyzed Ene-reactions, *Curr. Org. Chem.*, 2000, **4**, 30; (*e*) W. Adam and O. Krebs, The Nitroso Ene Reaction: a Regioselective and Stereoselective Allylic Nitrogen Functionalization of Mechanistic Delight and Synthetic Potential, *Chem. Rev.*, 2003, **103**, 4131. For a review on transition metal-catalyzed Alder–ene reactions, see: (*f*) B. M. Trost, M. U. Frederiksen and M. T. Rudd, Ruthenium–catalyzed Reactions—a Treasure Trove of Atom–economic Transformations, *Angew. Chem., Int. Ed.*, 2005, **44**, 6630.

2 N. G. Rondan, L. N. Domelsmith, K. N. Houk, A. T. Bowne and R. H. Levin, The Relative Rates of Electron-rich and Electron-deficient Alkene Cycloadditions to Benzyne. Enhanced Electrophilicity as a Consequence of Alkyne Bending Distortions, *Tetrahedron Lett.*, 1979, **20**, 3237.

3 For Alder–ene reactions of arynes, see: (*a*) I. Tabushi, K. Okazaki and R. Oda, Relative Reactivities of Substituted Olefins toward Benzyne, *Tetrahedron*, 1969, **25**, 4401; (*b*) G. Ahlgren and B. Akermark, The Addition of Benzyne to 1, 2-Dideuteriocyclohexene, *Tetrahedron Lett.*, 1970, **11**, 3047; (*c*) V. Garsky, D. F. Koster and R. T. Arnold, Studies of the Stereochemistry and Mechanism of the Ene Reaction using Specifically Deuterated Pinenes, *J. Am. Chem. Soc.*, 1974, **96**, 4207; (*d*) J. Nakayama and K. Yoshimura, A General Synthesis of Aromatic Compounds Carrying Two Neopentyl Groups on Adjacent Positions, *Tetrahedron Lett.*, 1994, **35**, 2709; (*e*) A. A. Aly, N. K. Mohamed, A. A. Hassan and A.-F. E. Mourad, Reaction of Diimines and Benzyne, *Tetrahedron*, 1999, **55**, 1111; (*f*) A. A. Aly and R. M. Shaker, 5-Benzyl-1H-tetrazols from the Reaction of 1-Aryl-5-methyl-1H-tetrazoles with 1, 2-Dehydrobenzene, *Tetrahedron Lett.*, 2005, **46**, 2679.

4 T. T. Jayanth, M. Jeganmohan, M.-J. Cheng, S.-Y. Chu and C.-H. Cheng, Ene Reaction of Arynes with Alkynes, *J. Am. Chem. Soc.*, 2006, **128**, 2232.

5 (*a*) Z. Chen, J. Liang, J. Yin, G. A. Yu and S. H. Liu, Alder-ene Reaction of Aryne with Olefins, *Tetrahedron Lett.*, 2013, **54**, 5785; (*b*) P. Pérez and L. R. Domingo, A DFT Study of Inter–and Intramolecular Aryne Ene Reactions, *Eur. J. Org. Chem.*, 2015, 2826.

 This journal is © the Partner Organisations 2021

Published on 15 April 2021. Downloaded by Fudan University on 1/18/2022 2:29:23 AM.

6 S. Gupta, P. Xie, Y. Xia and D. Lee, Reactivity and Selectivity in the Intermolecular Alder–Ene Reactions of Arynes with Functionalized Alkenes, *Org. Lett.*, 2017, **19**, 5162.

7 (*a*) T. R. Hoye, B. Baire, D. Niu, P. H. Willoughby and B. P. Woods, The Hexadehydro-Diels–Alder Reaction, *Nature*, 2012, **490**, 208; (*b*) B. Baire, D. Niu, P. H. Willoughby, B. P. Woods and T. R. Hoye, Synthesis of Complex Benzenoids via the Intermediate Generation of o-Benzynes through the Hexadehydro-Diels-Alder Reaction, *Nat. Protoc.*, 2013, **8**, 501; (*c*) D. Niu, P. H. Willoughby, B. Baire, B. P. Woods and T. R. Hoye, Alkane Desaturation by Concerted Double Hydrogen Atom Transfer to Benzyne, *Nature*, 2013, **501**, 53; (*d*) R. Karmakar, S. Ghorai, Y. Xia and D. Lee, Synthesis of Phenolic Compounds by Trapping Arynes with a Hydroxy Surrogate, *Molecules*, 2015, **20**, 15862; (*e*) T. Wang and T. R. Hoye, Hexadehydro-Diels–Alder (HDDA)-Enabled Carbazolyne Chemistry: Single Step, de Novo Construction of the Pyranocarbazole Core of Alkaloids of the Murraya koenigii (Curry Tree) Family, *J. Am. Chem. Soc.*, 2016, **138**, 13870; (*f*) F. Xu, X. Xiao and T. R. Hoye, Reactions of HDDA-derived Benzynes with Perylenes: Rapid Construction of Polycyclic Aromatic Compounds, *Org. Lett.*, 2016, **18**, 5636; (*g*) J. Chen, V. Palani and T. R. Hoye, Reactions of HDDA-derived Benzynes with Sulfides: Mechanism, Modes, and Three-component Reactions, *J. Am. Chem. Soc.*, 2016, **138**, 4318; (*h*) V. Palani, J. Chen and T. R. Hoye, Reactions of Hexadehydro-Diels–Alder (HDDA)-Derived Benzynes with Thioamides: Synthesis of Dihydrobenzothiazino-Heterocyclics, *Org. Lett.*, 2016, **18**, 6312; (*i*) R. Karmakar, P. Mamidipalli, R. M. Salzman, S. Hong, S. Y. Yun, Y. Xia and D. Lee, Benzannulation of Triynes Initiated by an Alder-ene Reaction and Subsequent Trifluoromethylthiolate Addition, *Org. Lett.*, 2016, **18**, 3530; (*j*) S. P. Ross and T. R. Hoye, Reactions of Hexadehydro-Diels–Alder Benzynes with Structurally Complex Multifunctional Natural Products, *Nat. Chem.*, 2017, **9**, 523; (*k*) S. Ghorai and D. Lee, Aryne Formation via the Hexadehydro Diels-Alder Reaction and Their Ritter-type Transformations Catalyzed by a Cationic Silver Complex, *Tetrahedron Lett.*, 2017, **73**, 4062; (*l*) Q. Hu, L. Li, F. Yin, H. Zhang, Y. Hu, B. Liu and Y. Hu, Fused Multifunctionalized Isoindole-1, 3-diones via the Coupled Oxidation of Imidazoles and Tetraynes, *RSC Adv.*, 2017, **7**, 49810; (*m*) X. Meng, S. Lv, D. Cheng, Q. Hu, J. Ma, B. Liu and Y. Hu, Fused Multifunctionalized Chromenes from Tetraynes and α, β–Unsaturated Aldehydes, *Chem. – Eur. J.*, 2017, **23**, 6264; (*n*) Y. Hu, J. Ma, L. Li, Q. Hu, S. Lv, B. Liu and S. Wang, Fused Multifunctionalized Dibenzoselenophenes from Tetraynes, *Chem. Commun.*, 2017, **53**, 1542. For reviews, see: (*o*) O. J. Diamond and T. B. Marder, Methodology and Applications of the Hexadehydro-Diels–Alder (HDDA) Reaction, *Org. Chem. Front.*, 2017, **4**, 891; (*p*) R. Karmakar, A. Le, P. Xie, Y. Xia and D. Lee, Reactivity of Arynes for Arene Dearomatization, *Org. Lett.*, 2018, **20**, 4168; (*q*) S. Ghorai and D. Lee, Synthesis of Imides, Imidates, Amidines, and Amides by Intercepting the Aryne–Isocyanide Adduct with Weak Nucleophiles, *Org. Lett.*, 2019, **21**, 7390; (*r*) S. Ghorai, Y. Lin, Y. Xia, J. D. Wink and D. Lee, Silver-Catalyzed Selective Multicomponent Coupling Reactions of Arynes with Nitriles and Isonitriles, *Org. Lett.*, 2020, **22**, 642; (*s*) S. Ghorai, Y. Lin, Y. Xia, J. D. Wink and D. Lee, Silver-catalyzed Annulation of Arynes with Nitriles for Synthesis of Structurally Diverse Quinazolines, *Org. Lett.*, 2020, **22**, 626. For mechanistic and theoretical studies of HDDA reaction, see: (*t*) A. Ajaz, A. Z. Bradley, R. C. Burrell, W. H. H. Li, K. J. Daoust, L. B. Bovee, K. J. DiRico and R. P. Johnson, Concerted vs Stepwise Mechanisms in Dehydro-Diels–Alder Reactions, *J. Org. Chem.*, 2011, **76**, 9320; (*u*) P. H. Willoughby, D. Niu, T. Wang, M. K. Haj, C. J. Cramer and T. R. Hoye, Mechanism of the Reactions of Alcohols with o-Benzynes, *J. Am. Chem. Soc.*, 2014, **136**, 13657; (*v*) B. Baire, T. Wang and T. R. Hoye, Tactics for Probing Aryne Reactivity: Mechanistic Studies of Silicon–oxygen Bond Cleavage During the Trapping of (HDDA-generated) Benzynes by Silyl Ethers, *Chem. Sci.*, 2014, **5**, 545; (*w*) Y. Liaung, X. Hong, P. Yu and K. N. Houk, Why Alkynyl Substituents Dramatically Accelerate Hexadehydro-Diels–Alder (HDDA) Reactions: Stepwise Mechanisms of HDDA Cycloadditions, *Org. Lett.*, 2014, **16**, 5702; (*x*) D. J. Marell, L. R. Furan, B. P. Woods, X. Lei, A. J. Bendelsmith, C. J. Cramer, T. R. Hoye and K. T. Kuwata, Mechanism of the Intramolecular Hexadehydro-Diels–Alder Reaction, *J. Org. Chem.*, 2015, **80**, 11744; (*y*) T. Wang, D. Niu and T. R. Hoye, The Hexadehydro-Diels–Alder Cycloisomerization Reaction Proceeds by a Stepwise Mechanism, *J. Am. Chem. Soc.*, 2016, **138**, 7832.

8 V. R. Sabbasani, G. Huang, Y. Xia and D. Lee, Facile Alder–Ene Reactions of Silylallenes Involving an Allenic C (sp^2)–H Bond, *Chem. – Eur. J.*, 2015, **21**, 17210.

9 (*a*) D. A. Candito, J. Panteleev and M. Lautens, Intramolecular Aryne–ene Reaction: Synthetic and Mechanistic Studies, *J. Am. Chem. Soc.*, 2011, **133**, 14200; (*b*) D. A. Candito, D. Dobrovolsky and M. Lautens, Development of an Intramolecular Aryne-ene Reaction and Application to the Formal Synthesis of (±)-Crinine, *J. Am. Chem. Soc.*, 2012, **134**, 15572.

10 (*a*) D. Niu and T. R. Hoye, The Aromatic Ene Reaction, *Nat. Chem.*, 2014, **6**, 34; (*b*) J. Zhang, D. Niu, V. A. Brinker and T. R. Hoye, The Phenol–ene Reaction: Biaryl Synthesis via Trapping Reactions between HDDA-generated Benzynes and Phenolics, *Org. Lett.*, 2016, **18**, 5596.

11 (*a*) R. Karmakar, P. Mamidipalli, S. Y. Yun and D. Lee, Alder-ene Reactions of Arynes, *Org. Lett.*, 2013, **15**, 1938; (*b*) R. Karmakar, S. Y. Yun, Y. Chen, Y. Xia and D. Lee, Benzannulation of Triynes to Generate Functionalized Arenes by Spontaneous Incorporation of Nucleophiles, *Angew. Chem., Int. Ed.*, 2015, **54**, 6582; (*c*) S. Gupta, P. Xie, Y. Xia and D. Lee, Reactivity and Selectivity in the Intermolecular Alder–Ene Reactions of Arynes with Functionalized Alkenes, *Org. Lett.*, 2017, **19**, 5162;

Published on 15 April 2021. Downloaded by Fudan University on 1/18/2022 2:29:23 AM.

(*d*) S. Gupta, P. Xie, Y. Xia and D. Lee, Reactivity of Arynes toward Functionalized Alkenes: Intermolecular Alder-ene vs. Addition Reactions, *Org. Chem. Front.*, 2018, **5**, 2208; (*e*) S. Gupta, Y. Lin, Y. Xia, J. D. Wink and D. Lee, Alder-ene Reactions Driven by High Steric Strain and Bond Angle Distortion to Form Benzocyclobutenes, *Chem. Sci.*, 2019, **10**, 2212. For examples of intramolecular Alder–ene reactions between arynes and alkenes, see: (*f*) H. Xu, J. He, J. Shi, L. Tan, D. Qiu, X. Luo and Y. Li, Domino Aryne Annulation via a Nucleophilic-Ene Process, *J. Am. Chem. Soc.*, 2018, **140**, 3555.

12 For Alder–ene reactions of allenes, see: (*a*) S.-H. Dal and W. R. Dolber Jr., Allene-perfluorocyclobutanone Ene reaction. Kinetic Isotope Effects, *J. Am. Chem. Soc.*, 1972, **94**, 3953; (*b*) C. B. Lee and D. R. Taylor, Ene Reactions of Allenes. Part 3. Reactions of Electron-deficient Azo-compounds with Acyclic Allenes and Alkenes, *J. Chem. Soc., Perkin Trans. 1*, 1977, 1463; (*c*) K. M. Brummond, H. Chen, P. Sill and L. You, A Rhodium(I)-catalyzed Formal Allenic Alder Ene Reaction for the Rapid and Stereoselective Assembly of Cross-conjugated Trienes, *J. Am. Chem. Soc.*, 2002, **124**, 15186; (*d*) P. H.-Y. Cheong, P. Morganelli, M. R. Luzung, K. N. Houk and F. D. Toste, Gold-catalyzed Cycloisomerization of 1, 5-Allenynes via Dual Activation of an Ene Reaction, *J. Am. Chem. Soc.*, 2008, **130**, 4517.

13 For examples of thermal [2 + 2] cycloadditions of allenes with alkynes, see: (*a*) K. M. Brummond and D. Chen, Microwave-assisted Intramolecular [2 + 2] Allenic Cycloaddition Reaction for the Rapid Assembly of Bicyclo [4.2. 0] octa-1, 6-dienes and Bicyclo [5.2. 0] nona-1, 7-dienes, *Org. Lett.*, 2005, **7**, 3473; (*b*) C. Mukai, Y. Hara, Y. Miyashita and F. J. Inagaki, Thermal [2 + 2] Cycloaddition of Allenynes: Easy Construction of Bicyclo [6.2.0]deca-1,8-dienes, Bicyclo[5.2.0]nona-1,7-dienes, and Bicyclo[4.2.0]octa-1,6-dienes, *Org. Chem.*, 2007, **72**, 4454; (*c*) X. Jiang and S. Ma, Intramolecular [2 + 2]-Cycloaddition of Propargylic 2, 3-Allenoates for the Efficient Synthesis of 3-Oxabicyclo [4.2. 0] octa-1 (8), 5-dien-4-ones: a Dramatic Substituent Effect, *Tetrahedron Lett.*, 2007, **63**, 7589; (*d*) T. V. Ovaska and R. E. Kyne, Intramolecular Thermal Allenyne [2 + 2] Cycloadditions: Facile Construction of the 5–6–4 Ring Core of Sterpurene, *Tetrahedron Lett.*, 2008, **49**, 376; (*e*) M. R. Siebert, J. M. Osbourn, K. M. Brummond and D. J. Tantillo, Differentiating Mechanistic Possibilities for the Thermal, Intramolecular [2 + 2] Cycloaddition of Allene– Ynes, *J. Am. Chem. Soc.*, 2010, **132**, 11952; (*f*) W. Ding and N. Yoshikai, *Angew. Chem., Int. Ed.*, 2019, **58**, 2500.

14 For the preparation of *gem*-dimethyl allenes, see: (*a*) M. W. Wright, T. L. Smalley Jr., M. E. Welker and A. L. Rheingold, Synthesis of Cobalt Substituted 1,3-Diene Complexes with Unusual Structures and their Exo Selective Diels-Alder Reactions, *J. Am. Chem. Soc.*, 1994, **116**, 6777; (*b*) M. Murakami, S. Kadowaki and T. Matsuda, Molybdenum-catalyzed Ring-closing Metathesis of Allenynes, *Org. Lett.*, 2005, **7**, 3953; (*c*) A. Boutier, C. Kammerer-Pentier, N. Krause, G. Prestat and G. Poli, Pd–Catalyzed Asymmetric Synthesis of N–Allenyl Amides and Their Au–Catalyzed Cycloisomerizative Hydroalkylation: A New Route Toward Enantioenriched Pyrrolidones, *Chem. – Eur. J.*, 2012, **18**, 3840; (*d*) A. S. Kulandai Raj, B. S. Kale, B. D. Mokar and R.-S. Liu, Gold-Catalyzed N, O-Functionalizations of 6-Allenyl-1-ynes with N-Hydroxyanilines To Construct Benzo [b]-azepin-4-one Cores, *Org. Lett.*, 2017, **19**, 5340.

15 For the preparation of a terminal allene, see: (*a*) D. F. Maynard and W. H. Okamura, π-Electrocyclization of 1-Azatrienes to 1,2-Dihydropyridines, *J. Org. Chem.*, 1995, **60**, 1763; (*b*) S. Ma and W. Gao, Efficient Synthesis of 4-(2′-Alkenyl)-2, 5-dihydrofurans and 5, 6-Dihydro-2 H-pyrans via the Pd-Catalyzed Cyclizative Coupling Reaction of 2, 3-or 3, 4-Allenols with Allylic Halides, *J. Org. Chem.*, 2002, **67**, 6104.

16 For the preparation of a 1,3-disubstituted allene, see: D. J. Lippincott, R. T. H. Linstadt, M. R. Maser and B. H. Lipshutz, Synthesis of Functionalized [3],[4],[5] and [6] Dendralenes through Palladium–Catalyzed Cross–Couplings of Substituted Allenoates, *Angew. Chem., Int. Ed.*, 2017, **56**, 847.

17 For the preparation of a tetrasubstituted allene, see: D. Posevins, Y. Qiu and J.-E. Backvall, Highly Diastereoselective Palladium-catalyzed Oxidative Carbocyclization of Enallenes Assisted by a Weakly Coordinating Hydroxyl Group, *J. Am. Chem. Soc.*, 2018, **140**, 3210.

18 For the Diels–Alder reaction of arynes with benzene/or other aromatics to form benzobarrelenes, see: (*a*) G. Wittig and L. Pohmer, Intermediäre Bildung von Dehydrobenzol (Cyclohexa–dienin), *Angew. Chem.*, 1955, **67**, 348; (*b*) I. Tabushi, H. Yamada, Z. Yoshida and R. Oda, Reactions of Benzyne with Substituted Benzenes, *Bull. Chem. Soc. Jpn.*, 1977, **50**, 285; (*c*) J. Chen, B. Baire and T. R. Hoye, Cycloaddition Reactions of Azide, Furan, and Pyrrole units with Benzynes Generated by the Hexadehydro-Diels–Alder (HDDA) Reaction, *Heterocycles*, 2014, **88**, 1191; (*d*) D. Niu, T. Wang, B. P. Woods and T. R. Hoye, Dichlorination of (Hexadehydro-Diels–Alder Generated) Benzynes and a Protocol for Interrogating the Kinetic Order of Bimolecular Aryne Trapping Reactions, *Org. Lett.*, 2014, **16**, 254; (*e*) V. D. Pogula, T. Wang and T. R. Hoye, Intramolecular [4 + 2] Trapping of a Hexadehydro-Diels–Alder (HDDA) Benzyne by Tethered Arenes, *Org. Lett.*, 2015, **17**, 856; (*f*) Y. Wang and T. R. Hoye, Intramolecular Capture of HDDA-Derived Benzynes: (i) 6-to 12-Membered Ring Formation,(ii) Internally (vis-à-vis Remotely) Tethered Traps, and (iii) Role of the Rate of Trapping by the Benzynophile, *Org. Lett.*, 2018, **20**, 88.

19 For the reactivity difference between triynes and tetrayenes in the hexadehydro Diels–Alder reaction, see: (*a*) A. Z. Bradley and R. P. Johnson, Thermolysis of 1, 3,

 This journal is © the Partner Organisations 2021

8-Nonatriyne: Evidence for Intramolecular [2 + 4] Cycloaromatization to a Benzyne Intermediate, *J. Am. Chem. Soc.*, 1997, **119**, 9917; (*b*) K. Miyawaki, R. Suzuki, T. Kawano and I. Ueda, Cycloaromatization of a non-conjugated polyenyne system: Synthesis of 5H-Benzo [d] fluoreno [3, 2-b] pyrans via Diradicals Generated from 1-[2-{4-(2-Alkoxymethylphenyl) butan-1, 3-diynyl}] phenylpentan-2, 4-diyn-1-ols and Trapping Evidence for the 1, 2-Didehydrobenzene Diradical, *Tetrahedron Lett.*, 1997, **38**, 3943.

Published on 15 April 2021. Downloaded by Fudan University on 1/18/2022 2:29:23 AM.